Universal gravitation

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Predict how gravitational force changes with mass and separation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 1.*
Changing heading or a curved sequence of positions shows direction changes; velocity includes direction.

02. Worked example: explain the model
Center separation r then 2 r; each body's force drops from 20 to 5 N, with equal opposing forces within each pair.

03. Read and interpret the model
Distance is between centers; equal opposite attractive forces.

04. Apply the physics
24 N; 3 N.

05. Explain your reasoning
Equal forces; smaller moon usually has greater acceleration from that interaction.

06. Review: connect an earlier idea *Review: Unit 1, Day 14.*
Ground pushes the runner’s foot forward with 60 N. This external force can accelerate the runner; the partner forces act on different objects.

07. Exit check
2 N.

08. STRETCH: test the model
12×2/4=6 N; increased mass partly offsets the greater separation.

Universal gravitation

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Predict how gravitational force changes with mass and separation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 1.*
Direction changes but speed remains 5 m/s. Instantaneous perpendicular force does no work; velocity is not constant.

02. Worked example: explain the model
Center separation r then 2 r; each body's force drops from 20 to 5 N, with equal opposing forces within each pair.

03. Read and interpret the model
Distance is between centers; equal opposite attractive forces.

04. Apply the physics
$3\times 2/3^2=2/3$.

05. Explain your reasoning
Equal forces; smaller moon usually has greater acceleration from that interaction.

06. Review: connect an earlier idea *Review: Unit 1, Day 14.*
Equal opposite 24 N forces. Craft acceleration 6 m/s^2; asteroid 0.06 m/s^2 opposite. Compares this interaction’s contribution, or assumes other forces negligible.

07. Exit check
3 r for unchanged masses. If both triple, need 9 r to reach $F_0/9$.

08. STRETCH: test the model
Factor 9 in numerator: separation 3 r preserves force; $3\sqrt{2}$ r gives half force.